

Lab06_FernandaVincenzo

Fernanda Di Vincenzo 280867

Pacotes

```
library(RSQLite)
```

Warning: pacote 'RSQLite' foi compilado no R versão 4.5.3

```
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

```
library(stringr)
```

1. Caminho do arquivo disco.db

```
fname <- file.path("datasets", "disco.db")  
fname
```

```
[1] "datasets/disco.db"
```

```
file.exists(fname)
```

```
[1] TRUE
```

2. Conexão com o banco de dados

```
conn <- dbConnect(SQLite(), fname)
conn
```

```
<SQLiteConnection>
  Path: C:\Users\nanda\Documents\ME315\datasets\disco.db
  Extensions: TRUE
```

3. Tabelas existentes no banco

```
dbListTables(conn)
```

```
[1] "albums"          "artists"          "customers"         "employees"
[5] "genres"          "invoice_items"    "invoices"          "media_types"
[9] "playlist_track"  "playlists"        "sqlite_sequence"  "sqlite_stat1"
[13] "tracks"
```

4. Colunas da tabela customers

```
dbListFields(conn, "customers")
```

```
[1] "CustomerId"  "FirstName"  "LastName"  "Company"  "Address"
[6] "City"        "State"      "Country"   "PostalCode" "Phone"
[11] "Fax"         "Email"      "SupportRepId"
```

5. Número de clientes cadastrados

```
dbGetQuery(conn, "SELECT COUNT(*) AS n_clientes FROM customers")
```

```
      n_clientes  
1           59
```

6. Número de países diferentes

```
dbGetQuery(conn, "SELECT COUNT(DISTINCT country) AS n_paises FROM customers")
```

```
      n_paises  
1           24
```

7. Clientes por país (ordem decrescente)

```
sql <- paste("SELECT country, COUNT(*) AS n_clientes",  
             "FROM customers",  
             "GROUP BY country",  
             "ORDER BY n_clientes DESC")  
clientes_pais <- dbGetQuery(conn, sql)  
clientes_pais
```

	Country	n_clientes
1	USA	13
2	Canada	8
3	France	5
4	Brazil	5
5	Germany	4
6	United Kingdom	3
7	Portugal	2
8	India	2
9	Czech Republic	2
10	Sweden	1
11	Spain	1
12	Poland	1
13	Norway	1
14	Netherlands	1

15	Italy	1
16	Ireland	1
17	Hungary	1
18	Finland	1
19	Denmark	1
20	Chile	1
21	Belgium	1
22	Austria	1
23	Australia	1
24	Argentina	1

8. Os 5 países com mais clientes

```
sql <- paste("SELECT country, COUNT(*) AS n_clientes",
             "FROM customers",
             "GROUP BY country",
             "ORDER BY n_clientes DESC",
             "LIMIT 5")
dbGetQuery(conn, sql)
```

	Country	n_clientes
1	USA	13
2	Canada	8
3	France	5
4	Brazil	5
5	Germany	4

:::

9. Países cujo nome possui exatamente 6 letras

```
sql <- "SELECT DISTINCT country FROM customers WHERE country GLOB '?????'"
dbGetQuery(conn, sql)
```

	Country
1	Brazil
2	Canada
3	Norway

```
4 France
5 Poland
6 Sweden
```

Conferência com dplyr e stringr:

```
clientes_pais |>
  filter(str_length(Country) == 6) |>
  select(Country)
```

```
Country
1 Canada
2 France
3 Brazil
4 Sweden
5 Poland
6 Norway
```

10. Músicas compradas por clientes brasileiros

```
sql <- paste(
  "SELECT DISTINCT tracks.trackid, tracks.name",
  "FROM customers",
  "INNER JOIN invoices      ON invoices.customerid = customers.customerid",
  "INNER JOIN invoice_items ON invoice_items.invoiceid = invoices.invoiceid",
  "INNER JOIN tracks        ON tracks.trackid = invoice_items.trackid",
  "WHERE customers.country = 'Brazil'"
)
musicas_br <- dbGetQuery(conn, sql) |>
  as_tibble() |>
  arrange(Name)

musicas_br
```

```
# A tibble: 190 x 2
  TrackId Name
  <int> <chr>
1    2190 1/2 Full
2    1345 2 Minutes To Midnight
```

```
3      298 A Cor Do Sol
4     1731 A Cura
5     1767 A Menina Dança
6     1785 Abraham, Martin And John
7     1344 Aces High
8      738 Admirável Gado Novo
9     2991 All Along The Watchtower
10    3017 All I Want Is You
# i 180 more rows
```

```
nrow(musicas_br)
```

```
[1] 190
```

11. Desconectar do banco de dados

```
dbDisconnect(conn)
```

Horário de Compilamento

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-20 21:28:11 -03"
```